EchoBreaker: How People Form and Update Beliefs About Public Opinion During Social Media Browsing

Kyu Lee (kbl2140)

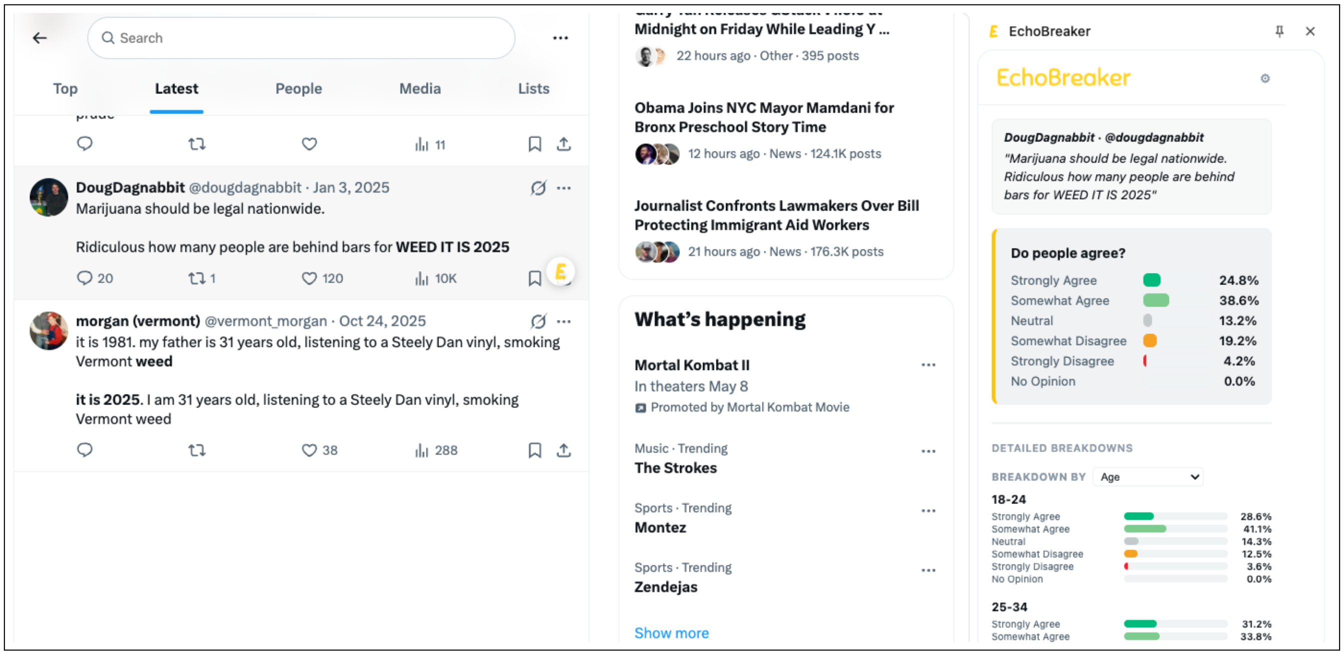

Figure 1: Using EchoBreaker. When a user hovers over a post on X (left), an EchoBreaker icon appears in the bottom-right corner of the tweet (as shown on @DougDagnabbit's post). Clicking the icon triggers a synthetic survey on the claim, and the results are displayed in the EchoBreaker sidebar (right), showing aggregate agreement levels on a Likert scale along with detailed breakdowns across demographic groups such as age.

1 Overview

Over the past couple of decades, social media platforms have increasingly become a major source of content consumption, and for some users, the dominant source. As a result, social media plays a major role in how people perceive how their fellow citizens think [17, 20, 23]. Imagine a user browsing through X, formally known as Twitter. As they scroll, the user sees a stream of opinion statements and factual claims, each accompanied by contextual data that shapes how the post lands. This contextual information includes engagement metrics (likes, views, reposts, reply counts, etc.), the author of the post, the tone of the most engaged replies, the general sentiment of replies, whether other independent posts share the viewpoint, and whether the post was viewed as a result of an algorithm, among many others. Through this process, users construct impressions of public opinion, which research has shown to be different than actual public opinion, most often measured by polling data. The difference between perceived and actual public opinion is called the perception gap.

Most users are intellectually aware that what appears on social media is often not representative of the sentiments of the general public. They know that the loudest voices usually win, and that algorithmic feeds are curated to optimize for engagement, which often means promoting posts designed to elicit the strongest emotional responses. A motivation behind this paper, therefore, is not to study whether people mistake social media sentiment for general public sentiment, but to understand what and how contextual information influences a users' intuitive estimates of public opinion, despite their intellectual understanding that social media is not a poll. A useful analogy is the Müller-Lyer illusion, a well-studied optical illusion in which lines of identical length appear to be different. (see Figure 2)

Human beings intellectually know that the lines are of identical length; after all, they can literally measure it with a ruler. However, even after measurement, people continue to perceive one as longer than the others and act accordingly.

On its own, perception gaps might seem inconsequential; however, these public opinion distortions do matter because they can influence actual human behavior. For example, perception gaps can produce pluralistic ignorance, a well-studied phenomenon whose core idea can be defined by: "No one believes, but everyone believes that everybody else believes." In essence, social media users might believe their opinion is a minority, even if it is the majority, due to content on a platform's feed. This causes them to be silent and the distorted public sentiment is reinforced for everyone else [25]. What may follow include alienation from society, ill-placed contempt for fellow citizens, erosion of trust in various institutions, unreasonable distrust for "the other side," and radicalization to create a sense of belonging, among many others [34]. On the other hand, false consensus effect, a tendency to overestimate the public support for one's own views, can also have real-world behavioral implications. Especially when people with fringe views have false consensus (often via social media activity), they can become emboldened to further engage in activities that perpetuate their views and ideas.

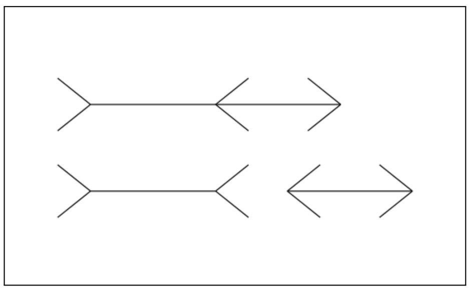

Figure 2: All the lines here are of identical length.

There has been ample literature studying which topics are most susceptible to perception gaps and how wide they are; this is usually quantified by asking research participants to estimate public opinion on some topic (e.g. politics, technology, lifestyle) and then comparing it to actual polling data [3]. In addition, there has been a substantial amount of research on misinformation within social media and how to combat it, often with various fact-checking tools. While both bodies of work are greatly valuable, this paper asks two different, but related questions: first, how do perception gaps form in real-time within the specific context of a specific social media post? How much do engagement metrics and content of replies influence these gaps versus one's own pre-existing opinions? Note that this is distinct from studying how contextual information impacts one's own viewpoints, a phenomenon that is well researched [10, 13, 26]. Second, what happens when users are confronted with potential contradictory data exactly at the gap formation stage? Do they update their beliefs? Are they reflective or combative?

To address these questions, we introduce EchoBreaker, a web browser extension that lets users select specific social media posts (tweets, reddit posts/comments, youtube comments, etc.) and shows you a simulated public sentiment breakdown using LLM synthetic surveys. While to our best knowledge, EchoBreaker is the first tool of its kind, it is not the main contribution of this paper. Rather, EchoBreaker is a research artifact to enable our study. Also, it is worth noting that this research project is intended for the HCI community, not strictly for the Machine Learning or Software Engineering community. So the ways we "evaluate" EchoBreaker might seem different. Instead of doing some sort of evaluation against an existing benchmark, we are trying to understand what are the psychological and human behavioral learnings we can extract from people using EchoBreaker.

Furthermore, we plan to continue pursuing doing research with EchoBreaker after Dr. Kaiser's 6156 course. For the full research paper, we plan to run a two-part mixed-methods study: a longitudinal observation (N = 10) followed by a pre-registered asynchronous survey (N = 200). But this is outside the scope of this class and would also require IRB approval. Therefore, for the purposes of this class, the "evaluation" or "study" is a pilot study (N=3), where we present EchoBreaker to the participants and observe their responses/feedback on the tool.

2 Github Repository

https://github.com/danielkyulee/echobreaker

Any target users (anyone who uses Twitter or Reddit regularly!) can clone the "extension" directory in this repository and install themselves on a Chrome browser. No need to run the backend as that part is already deployed on the cloud.

3 Definitions

The following important definitions were introduced and defined in the "Overview" section above. However, for clarity, we will also reiterate it here.

Contextual Information (in Social Media Post): this includes all data and information that accompany a social media post. This includes (not comprehensively) the like count, the number of replies, the replies itself, the tone of the replies, in which feed the post appears (following or for you, etc.), the author of the post, the number of reposts, and how often it shows up among many others.

LLM Synthetic Survey: is a method in which one tries to mimic polling data by creating personas that reflect actual humans and then asking those personas what their opinion is on a specific topic.

Perception Gaps: the difference between a person's perception of what the public believes and what the public actually believes.

Pluralistic Ignorance: a psychological phenomenon in which people believe that everyone else believes in a particular opinion, even though not a lot of people actually believe in it.

4 Research Questions

RQ1 (Formation): What social media contextual data and what user characteristics predict the magnitude of the perception gap between a user's public opinion estimate and EchoBreaker's estimate?

Here, we examine what factors are most responsible in creating perception gaps at the moment that a user comes across a particular social media content. Is it related to engagement metrics such as like count or reply count that may signal public consensus? Or does already existing personal beliefs triumph all contextual data, or at least, to what degree?

RQ2 (Response): When users are confronted with EchoBreaker's data that contradicts their own public opinion estimates, how do they respond? Does the response evolve with repeated exposures over a prolonged period of time?

If EchoBreaker shows data that seems to conflict a user's perception of public opinion, how do users respond? Do they they update their beliefs, or do they combatively dismiss the tool? Perhaps they try to rationalize the discrepancy? We want to study these instances at the moment it happens (when a user views a specific post on social media) while also measuring whether the responses change over a prolonged period of time.

RQ3 (Design): What features of a real-time, in situ callibration tool (like EchoBreaker) impact a user's receptivity to it, especially when it conflicts with their own perceptions?

Is the way that EchoBreaker presents its data hindering user's acceptance of it? How can we optimize the UX to better address perception gaps in social media? Our goal is to provide design guidelines based on our study to help future researchers build upon our work.

5 The System: How Does EchoBreaker Work?

EchoBreaker is a Chrome browser extension that can be installed just like any other extensions. It has to be utilized on desktop and is not compatible with the mobile experience.

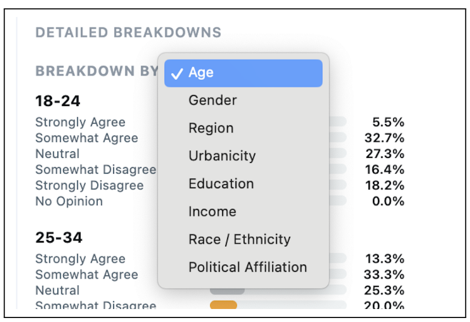

Figure 3: Demographic dimensions users can break the data down by.

Even though EchoBreaker can be used on X/Twitter and Reddit, for the purposes of this paper, we will do our study with participants only on X/Twitter.

While a user is browsing social media (like X/Twitter), a user can hover their mouse over a specific post which will trigger the Echo-Breaker icon to pop up on the bottom left of the post. Upon clicking the icon, EchoBreaker's interface will expand on the right side of the browser (a typical browser extension behavior) and start running an LLM synthetic survey on that specific post. For example, let's say a user comes across a tweet that reads "We should have a national minimum wage of 20 dollars per hour" by someone with the handle "@dandan12345". EchoBreaker will then proceed to survey all of its simulated participants and ask it to pick one of the following choices based on the tweet: strongly agree, somewhat agree, neutral, somewhat disagree, strongly disagree, and no opinion. After running the simulated polling, Echo-Breaker will present the aggregated results to the user. The user can segment the data by various dimensions such as age, gender, political affiliation, urbancity, education, and income (see Figure 3). To see a video demo of EchoBreaker in use, view this link: https://www.loom.com/share/d5e5b020c8cb4cda8385083a3f869482.

6 Study Design

To answer my research questions, I recruited three participants and interviewed them for 45 minutes each. I followed the following order and line of questioning.

- Establish Baseline: I asked the following three questions.
  - When you scroll through Twitter/X, how much do you feel like the opinions you see reflect what most people actually think?
  - Intellectually, do you think what you see on [platform] is representative of the general public? And does that knowledge actually change how things feel when you're scrolling?
  - How much do you trust surveys or polls in general? Has that changed over time?
- Use EchoBreaker on 3 pre-selected tweets (see Figure 4) and have them use EchoBreaker on it. These following questions would be asked:
  - (BEFORE SEEING ECHOBREAKER RESULTS) How do you think the American public would respond to this statement? (strongly agree, somewhat agree, neutral, somewhat disagree, strongly disagree, no opinion — summing to 100%)
  - (BEFORE SEEING ECHOBREAKER RESULTS) What about what you're seeing here makes you think that?
  - (AFTER SEEING ECHOBREAKER RESULTS) How surprised are you by the results?
  - (AFTER SEEING ECHOBREAKER RESULTS) Do you trust Echobreaker's results or your own instinct more? Explain.
  - (AFTER SEEING ECHOBREAKER RESULTS) Did this change your belief on the topic? Explain.
  - (AFTER SEEING ECHOBREAKER RESULTS) Any other thoughts or feelings?
- To conclude the interview, I would ask the following final questions:
  - If you were using this tool over a long period of time, do you think it would change you in any way? Your opinions, your emotions, anything.
  - Was there anything confusing about the interface or the results — anything where you weren't sure what you were looking at?
  - Do you have any suggestions on how to improve the product? Anything that would make it more helpful for you?

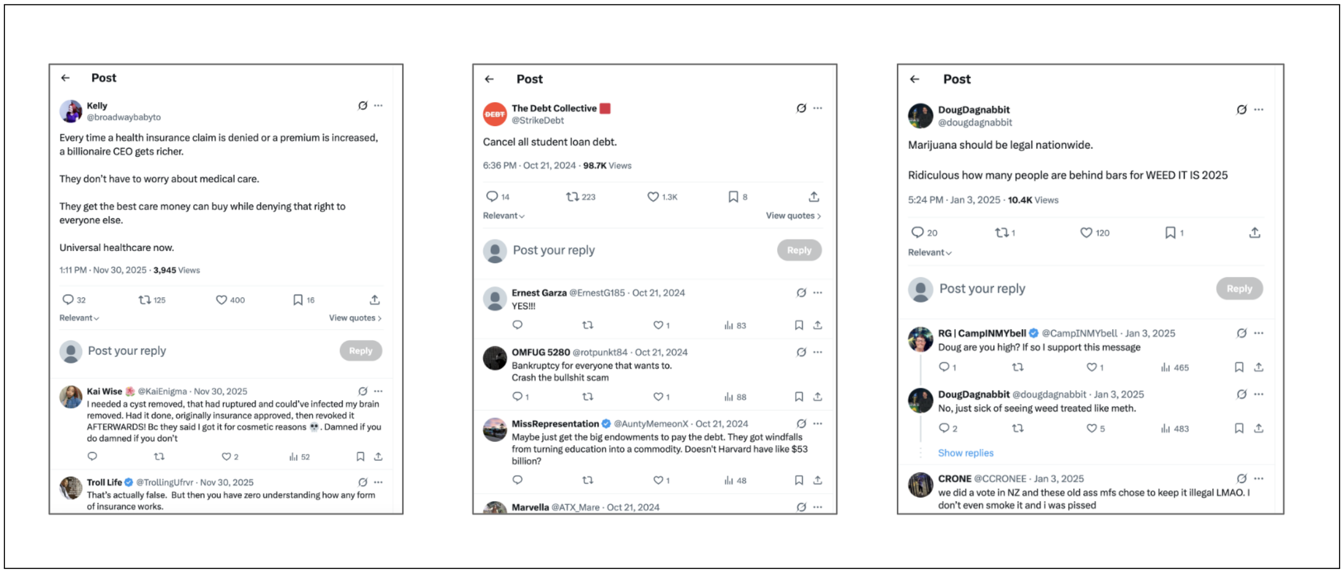

Figure 4: These were the 3 pre-selected tweets shown to participants.

Described below are the three participants for this study:

- P1 was between 40-44 years old, leaned democrat, used twitter a couple times per week. This participant almost rarely posts on social media. This person did not feel like the opinions on social media reflect what most people think, however, he mentioned that it "really depends on the topic and people you are reading from." He is also very trusting of polling data in general. This participant was recruited from Prolific.
- P2 was between 35-39 years old, has no political affiliation, used twitter almost every day. This participant never posts on social media. This person definitely does not feel like the opinions on social media reflect general population in any ways. She understood that only the most emotionally charged posts get populated by the algorithm. Furthermore, only a specific, minority group are even interested in posting on social media. She is also generally trusting of polling data in general. This participant was recruited from my network.
- P1 was between 30-34 years old, leaned republican, used twitter a couple times per week. This participant almost never posts on social media. This person also definitely does not feel like social media is representative of public opinion. He is trusting of general polling data. This participant was recruited from my network.

7 Study Findings

7.1 Regarding what contextual information impacts a user's perception of public opinion...

7.1.1 Engagement metrics (like count, reply count, repost count) do not play any meaningful role in a user's perception of public opinion. For example, if a post that read "America should have universal healthcare" had a significant number of likes (i.e. 10,000), it would not impact how the user perceives public opinion on the statement. It would not make them feel like the general American population particularly agrees or disagrees.

7.1.2 The general tone of the overall replies on a post only plays a marginal role in a user's perception of public opinion. For example, if the same post that read "America should have universal healthcare" had 1,000 replies and each reply was in favor of the statement...it would still not make the user believe that the general public's opinion would be favorable. The reason is because they realize that social media can often be an echo chamber.

7.1.3 The user's pre-existing perception on a statement from a particular post was pretty much the same before and after seeing the social media post and its contextual information. In other words, the contextual information on a post does not seem to meaningfully change or sway a user's perception of public opinion.

7.2 What do users do when presented with data that contradicts their perception of public opinion?

7.2.1 To my somewhat surprise, every participant in this particular study was actually really receptive to the results of EchoBreaker. If EchoBreaker presented data that was different from their own initial perceptions, each participant was quick to update their own beliefs. They were all reflective and acknowledged that they might have had misconceptions.

7.2.2 Furthermore, every participant trusted EchoBreaker, even though they understand the concept (and potential hallucinations) of LLM synthetic surveys. Even though each participant understood that EchoBreaker could be off, they thought the chances that they themselves were more off was much higher. Anecdotally, one of the participants

"It is more important to control gun violence than it is to protect gun rights"			
	Monmouth Polling	EchoBreaker (500 Personas)	EchoBreaker (1000 Personas)
Strongly Agree		15%	13%
Somewhat Agree	59%	27%	27%
Neutral	N/A	21%	24%
Somewhat Disagree		20%	19%
Strongly Disagree	40%	17%	17%
No Opinion	N/A	0%	0%

"We should deploy the National Guard into local communities to help reduce crime."			
	Monmouth Polling	EchoBreaker (500 Personas)	EchoBreaker (1000 Personas)
Strongly Agree	26%	17%	17%
Somewhat Agree	26%	25%	26%
Neutral	N/A	24%	26%
Somewhat Disagree	14%	26%	23%
Strongly Disagree	34%	8%	9%
No Opinion	N/A	0%	

Figure 5: Results of LLM synthetic survey evaluation.

mentioned that they "generally trusted LLMs" so they trusted EchoBreaker since it was based on the same LLM model. This indicates to me that there might be a correlation between trust in LLMs to trust in EchoBreaker.

7.2.3 However, I understand that this was a small study (N = 3) so to get more conviction in my conclusions, I would need to do a larger qualitative study.

7.3 Design learnings

7.3.1 I built a special "Research Version" of EchoBreaker to capture people's perception of public opinion BEFORE EchoBreaker shows its results so that I can capture that data so that I can measure the exact perception gap. Also, the "Research Version" asks users a series of questions after EchoBreaker shows its results such as "How surprised were you?" and "If the results changed your mind" so that I could study EchoBreaker's effects (See Figure 6). The intention of this feature was simply a way to efficiently gather necessary research data. However, to my big surprise, this feature actually played a huge role in helping achieve EchoBreaker's goal of closing the perception gap. Users essentially saw this feature as a game. They wanted to see if they could get the perception "correct." This did two things: first, it made EchoBreaker more enjoyable so they are more likely to use it. Second, if their answer was "off," it made them acknowledge that they were wrong and psychologically made them reflect on how they can improve themselves going forward. I realized that EchoBreaker is much more effective with this feature. This was perhaps the biggest single design learning I had and I believe it would be a big contribution to the UX community: gamification in a product like EchoBreaker can actually encourage more reflection.

7.3.2 Another learning was that the possible choices of "Strongly Agree", "Somewhat Agree", "Neutral", "Somewhat Disagree", and "Strongly Disagree" was not enough for a lot of users because it's possible they didn't have any opinion at all (or didn't even know what the post was about). I noticed this immediately based on the first participant interview and that is why I added a "No Opinion" option.

7.3.3 Although I am not sure this is a true "Design learning", i did notice that participants had a lot of trouble installing the Chrome Extension from a zip file that I had sent them. Also, participants seemed to be more hesitant to install for security and trust issues. Moving forward, I plan to launch the extension to the actual Chrome extension store so users can install directly from there.

8 Evaluation of LLM Synthetic Surveys

While the purpose of my research isn't to study the effectiveness of LLM synthetic surveys (there has already been a lot of research activity around this topic. See Related Works section), I still wanted to do a basic evaluation to see whether EchoBreaker's synthetic survey was at least directionally reliable. To do this, I found two statements with real polling data (https://maristpoll.marist.edu/polls/the-1st-amendment-in-the-u-s-october-2025/). Then I used EchoBreaker on a Twitter post with that exact same statement. I also used EchoBreaker with 500 simulated personas and 1000 simulated persons as my hypothesis is that the synthetic survey gets more accurate as there are more personas (because it gets more granular). The two tweet statements were "It is more important to control gun violence than it is to protect gun rights" and "We should deploy the National Guard into local communities to help reduce crime." (see Figure 5 for the results.)

From the evaluation, there are a few things you can observe: first, you can see that synthetic surveys are directionally correct, but the exact percentages are not exactly correct. This suggests that EchoBreaker could be reliable in simulating polling data in a more efficient and instantaneous manner. Second, the accuracy of the synthetic survey doesn't seem to increase with more personas. This doesn't necessarily mean that synthetic surveys can't be optimized, but that it might need more fine tuning apart from just increasing the number of personas.

It is also worth noting that the polling data to which we evaluated EchoBreaker didn't have the exact same options as EchoBreaker. for example, the polling does not give the "Neutral" or "No Opinion" option for participants. Furthermore, sometimes they do not distinguish between Strongly Agree and Somewhat Agree.

9 (Self Evaluation) What I Accomplished, What I Found Challenging, What I Learned

I was able to accomplish everything I said I would from my project proposal which included building the artifact itself and doing the pilot study (I actually originally planned for N = 2 but was able to interview 3 participants). In addition, I actually started writing the "research paper" which I plan to continue working on after Dr. Kaiser's course. This includes having thought through the more comprehensive two-phase mixed-methods along with the "related works" section. I put these two sections in the Appendix (although I'm not sure it belongs there) just to show what I have done because it does not really fit anywhere for the purposes of this course's final report. Note: I understand that a lot more work is required to refine the study design and related works section and it is something I will work on further after the conclusion of this class.

The most challenging aspect of my research was actually trying to come up with the right research questions and the right framing to make it interesting for the "Social Computing" group of the HCI community. At first, I wasn't sure if this was an artifact contribution or if it was an empirical study paper, or if this wasn't even an HCI paper at all (but instead a paper for like Psychology groups). But after talking with a few professors from Columbia and outside of Columbia who work on Social Computing, I'm starting to get a

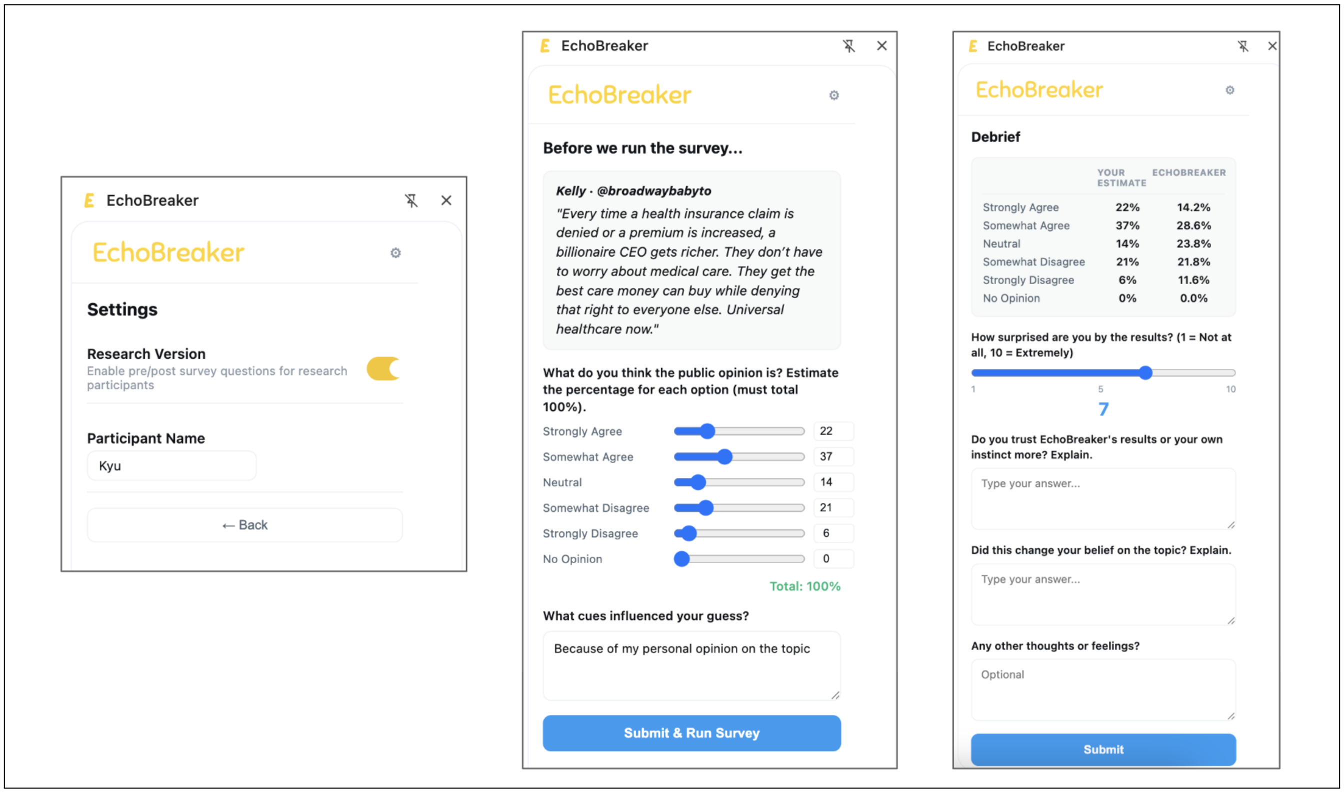

Figure 6: User can toggle on the "Research Version" in settings (left). With this enabled, whenever a user tries to run EchoBreaker, it will ask the user for their perception of public opinion before EchoBreaker shows its results (middle). After EchoBreaker shows its results, the user will also be asked to "Debrief" by asking various questions (right).

better idea of the how to frame this research to make it interesting for that community. Encouragingly, a lot of professors I have talked to think this project is interesting! In fact, I realized there are other professors from different institutions that are working on something very similar!

There is a lot I learned from this experience. Especially since I am new to research, everything was basically a new learning:

- Obviously, I learned a lot from doing the research study itself. I mentioned these in the "Study Findings" section so won't repeat it here.
- I sent each participant a zip file of my chrome extension for them to install. There was some friction here because some participants were worried that there was malware. Furthermore, one participant on an older Windows had trouble installing it. When I do the more comprehensive study, I should make sure to just launch it on the Chrome extension store to build trust and reduce technical errors. (I had deployed EchoBreaker's backend to GCP, but didn't launch the extension/front end).
- I tried to recruit participants with varied political affiliation through Prolific. While it was very easy to recruit participants who had no political affiliation, undisclosed affiliation, or democrat affiliation, it was harder to find a participant identified as republican. This is something I should keep in mind for the future.
- While writing the "Related Works" section, I realized it was going to take a lot more time for me to become an "expert" in those fields. I need to budget more time for this when I write the actual research paper.
- As part of this research project, I got to speak to a handful of professors in the HCI community. It was interesting to learn about how people in that community think of various ideas and the different processes each take to write papers. For example, I've noticed a pattern where some HCI professors spend a lot of time trying to frame the story from the very beginning; in contrast, other professors are more fluid and just "start talking to people" and then try to figure out the story along the way. These were all interesting learnings as I try to do HCI research as well!
- One of my concerns when starting this research was that the research community would get stuck on the fact that LLM synthetic surveys were not reliable. I did not want this to become a machine learning paper and synthetic surveys was simply a way to build a tool. After talking to a few professors, they taught me that it really depends on the framing of the paper. As long as I am able to show that previous research indicates it has

some reliability, it should be OK. Also, I need to make it clear that I do not expect synthetic surveys to be entirely accurate.

- I also learned more about the different HCI conferences/journals and all the subgroups within it.

10 Related Works

Our paper builds upon previous work from three different research areas: people's misperceptions of public opinion, how contextual information within social media content impacts user cognition, and simulating polling data using LLMs through synthetic surveys. While this "Related Works" section wasn't a requirement for Dr. Kaiser's 6156 course, I still included it because I thought it was highly relevant and so one can see the past research work I built off of. Graders do not have to read it if unnecessary.

10.1 Misperceptions of Public Opinion and its Consequences

There has been a lot of research activity around understanding the gap between people's perception of public opinion and actual broad public opinion. These "perception gaps" are often measured by asking study participants what they think public opinion is on a particular subject and then comparing it to existing polling data. Prior research has shown that people tend to view groups with opposing views as more extreme than they actually are, while also believing that groups they are aligned with are more unified than they actually are [2, 3, 15, 16, 21, 28].

This cognitive tendency is closely related to two well-researched social psychological phenomena: the false consensus effect and pluralistic ignorance. The false consensus effect is a psychological bias where people tend to think that others agree more than they actually do [24, 35]. Misperceptions regarding unity within aligned groups tend to further contribute to this effect because they believe their own groups will agree with them, whereas they will falsely categorize opposing groups as more extreme than reality [3, 16]. In contrast, pluralistic ignorance occurs when individuals mistakenly believe they are in the minority. As a result, they refrain from expressing their opinions to others and the wider public, which can reinforce and widen the perception gap [27, 32].

10.2 Correcting Perception Gaps

Given the societal implications of perception gaps, scholars have been researching how to effectively close these discrepancies. A perhaps obvious and intuitive method is to simply show people a more accurate data on what people really think; research indicates that in general, this works [3]. However, there are nuanced details that cannot be ignored. For example, a correction coming from someone you already align with (ie. someone from your same political affiliation) is more likely to close the perception gap than someone who is neutral or outside of your 'group' [7]. Also, the exact method in how you present a more accurate representation matters as well. One study by Kalla and colleagues showed that a more friendly, non-judgmental, face to face correction is far more effective in closing the perception gap [19].

Perhaps the most important question is not whether surfacing the right data can minimize perception gaps (it can), but how, when, and in what format to showcase this information for it to truly matter.

10.3 How Social Media Cues Impact Users' Attitudes and Behavior

A growing body of research indicates that the contextual information that goes alongside social media posts (engagement metrics, author authority, content of replies, whether author is in network, etc.) can shape how one processes and evaluates specific content. In other words, social media cues influence attitudes and behaviors. For example, it has been shown that social endorsements (when someone you know and trust likes or shares) matter more than whether one aligns politically with the publishing organization [26]. In addition, it seems that there is a direct correlation between policy attitudes and the number of likes and reposts on identical tweets [13].

Beyond simple engagement metrics and authorship, other contextual data, such as comments and replies, influence a users' cognition. In particular, emotionally charged and unruly comments underneath a post will do more to shift reader's perspective on an underlying issue [17].

While these work have focused on how these signals influence user's own attitudes, our research builds upon this to understand how impacts a user's perception of other people. Specifically, we study how various social media cues shape users' belief about what others believe, at the moment of seeing the post.

10.4 Distinguishing Perception Gaps from Misinformation

One nuance we want to clarify is that perception gaps are distinct from misinformation. Misinformation, a highly researched field also known as disinformation studies, is the study of how false information is created and spread, and subsequently influences people's understanding on a particular topic. As social media is often a contributor in spread of misinformation, it is crucial to understand how it happens and how to combat it.

Despite seeming similar, misinformation and perception gaps have a key distinction: the former is about accuracy about facts whereas the latter is about beliefs about public opinion. In fact, perception gaps can form and persistently exist even when individuals correctly understand that a particular statement is false. Our research focuses our efforts on understanding perception gaps within the context of social media.

10.5 Tools to Mitigate Social Media Distortions

To our best knowledge, no research studies have specifically looked into tools for mitigating perception gaps formed during social media browsing. However, there is a rather significant amount of work on combating other related social media distortions such as misinformation and algorithmic bias.

Regarding mitigating misinformation, a number of tools have been developed to help users evaluate whether a particular content is factually true or not. There are tools made by platforms themselves such as Community Notes on X/Twitter that leverage fact-checking through crowd-sourcing, which research indicates can be helpful [4, 11, 14]. Other tools enable users to verify the credibility of news articles and journal publications based on a variety of factors such as the team behind the publication and whether they have previously published false information. One such tool is called NewsGuard [22]. There are also browser extensions (like EchoBreaker) that enables users to fact-check specific online content, including on social media. Trustnet allows users to specify others who they trust to give an assessment [18] while Nudgecred uses 'design nudges' (color-coded indicators that represent credibility) based on source authority and crowd-sourced opinion [8]. These type interventions where a warning or label is attached does aid users in the realization of factually incorrect content. However, there can be unintended consequences such as users underestimating the correctness of factual pieces of content if they have no warnings or labels at all [12, 30]. This implies that there are many facets to misinformation mitigation designs.

Regarding algorithmic bias front, researchers have studied ways to let users know that feeds are curated by platforms who often prioritize for engagement, not necessarily for credibility. One way to create this awareness is to simply provide tools to surface content they otherwise would not have seen, or in other words, rerank the feeds [31]. The goal of these tools is of course to showcase ideologically diverse content which research suggests can reduce, though not eliminate, the partisan effects of social media's curated feeds [6].

10.6 How do LLM Synthetic Surveys Work and its Effectiveness

EchoBreaker uses LLM synthetic surveys to generate the public opinion data presented to users. The general idea is this: we create different human personas based on demographic data provided by the US census. For example, one such persona can be "37-year old, female, politically moderate, income of $85,000 per year, bachelor's degree...". We can also combine this census data with other polling data to create even more descriptive personas. With the simulated human personas, we can then essentially ask LLMs to predict what this persona would think about a particular content on social media (i.e. "The U.S. debt should be an important problem that politicians should address."). By repeating this prompt across many created personas, researchers can construct polling data, albeit 'synthetic', without actually polling real humans.

One of the first papers to introduce this concept was by Argyle and colleagues in their 2023 paper "Out of one, many: Using language models to simulate human samples". Here, the authors argue that "algorithmic bias" (which was and is still often considered a bug) that had been prevalent in GPT models could actually be used to accurately emulate real human responses from a variety of groups. They were the first to coin the term "silicon sampling" which essentially the act of creating numerous human personas with various demographics for surveying [5]. Various studies were published shortly after that showed that LLMs can, at least directionally, emulated human behavior when running simulated surveys [1, 29].

However, there exists research that seems to indicate a more skeptical view on the reliability of LLM synthetic surveys. In a 2024 paper by Bisbee and colleagues, the authors experimented with synthetic surveys using ChatGPT. They fed the LLM with various human personas and asked it to provide feeling thermometer scores regarding 11 sociopolitical groups. Their findings were that while the aggregated average results were fairly reliable, there is less variance in individual persona responses than the actual human data [9]. This is particularly important for EchoBreaker because the tool not only shows aggregate results, but also breaks down public opinion by more specific demographics such as age, gender, income, etc. So while the overall percentages might closely reflect real polling surveys, maybe it might not be as reliable when looking into more specific dimensions. Furthermore, Bisbee's paper also experimented with providing LLMs identical prompts across a period of 3 months and found that the results were significantly different. Other studies on LLM synthetic surveys found another concerning pattern: that large language models seem to misrepresent certain demographic groups, such as 65+ adults and widowed individuals [36], and that it performs better in Western countries, particular the United States [33].

Overall, current research indicates that LLM synthetic surveys, as it stands today, are a promising, yet imperfect mechanism. EchoBreaker understands this limitation and we do not claim that synthetic surveys is a replacement for real polling data. We do not position EchoBreaker as a tool for generating ground truth on public opinion, but rather as an artifact that can approximate real people to trigger a reflection with its users to mitigate perception gaps. Research has shown, after all, that provides a good approximation which is meaningful because perception gap reflection is unlikely when the gap is only marginally off. Also, given the immense real-world applications that arise from synthetic surveys, we predict synthetic surveys to garner significant research activity which will undoubtedly improve its reliability. The design guidelines we contribute will only become more meaningful as time goes on.

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A Study Design (After Class)

Note: I added this (along with Expected Contributions) to the Appendix to show what I will do after this course is over. This wasn’t a requirement for the final report so graders can skip if uninterested.

We will conduct a two-phase mixed-methods study to answer our research questions. Phase 1 (N=10) will be a qualitative, longitudinal observation and Phase 2 (N= 200) will be a pre-registered asynchronous survey. The purpose of Phase 1 is to discover patterns to help design the questions for Phase 2 so that we can test those patterns with more statistical significance.

Phase 0: While not the main part of our study, there are some prework that needs to be done before we can start Phase 1. First, we need to test the results of EchoBreaker against some real polling data (3 to 5). While LLM synthetic surveys won’t be 100% accurate (neither is polling), we need to be confident it is directionally correct. There are ways we can tweak EchoBreaker’s backend to improve the accuracy. Methods include iterating on the persons of simulated participants and increasing the number of simulated participants. Second, we need to ensure that EchoBreaker as a product is free from ’user-experience bugs.’ Is the product intuitive to use? Do the copy make sense to end users? We will do this by recruiting participants to utilize EchoBreaker and observe their usage and look for any potential friction points.

Phase 1: Longitudinal Study (N=10)

We will recruit 10 participants, likely through Prolific. These users must be frequent X/Twitter users (average once per day) and use the platform on desktop (they can be users of both mobile and desktop). We will ensure the final group of participants vary in political orientation, personal trust in polling data, intensity of social media use, age, and geographical location. Each participant in this phase will go through three different stages: Introductory session to kickstart study, 7 days where participants use EchoBreaker on their own time ("Diary Week"), and a closing interview to dig into patterns from their usage.

During the introductory session, we will install and introduce EchoBreaker to the participants. But more importantly, we will establish a baseline on whether they believe social media is reflective of the broader public population. This is important because we need to understand if any particular reactions during "Diary Week" are a result of the tool itself or the participants’ pre-existing skepticism of social media. Furthermore, we will get their first reactions of EchoBreaker’s survey on three pre-selected posts on X/Twitter. This will calibrate the participant to the task during "Diary Week," and also give us an opportunity to capture first impressions across all ten participants with the same experience.

During the 7-day "Diary Week," users will be instructed to continue their normal social media behavior with the only caveat being that they must trigger EchoBreaker at least two times per day. Each time, participants will be asked to the following questions before EchoBreaker reveals its survey results:

- What do you think the public opinion is? (Participants will estimate the percentage for each option with options being strongly agree, somewhat agree, neutral, somewhat disagree, strongly disagree, and no opinion. The participants will answer using a slider interface and the total must add up to 100%.)
- What cues influenced your guess? (This will be in text answer form.)

After seeing the synthetic survey results, participants will then be asked to debrief by answering the following questions:

- How surprised are you by the results?
- Do you trust EchoBreaker’s results or your own instinct more? Explain.
- Did this change your belief on this topic? Explain.
- Any other thoughts or feelings?

All of above questions will be integrated directly into the user flow of EchoBreaker as long as participants toggle on "Research Version" within the Settings. Participants from Phase 1 will be instructed to do so during the introductory session.

After 7 days, we will conduct a closing interview where we reflect on the participants experiences and dig further into the patterns discovered. The analysis here will help us form the hypothesis for Phase 2.

Phase 2: Asynchronous Survey (N = 200)

During Phase 2, we will recruit approximately 200 participants (likely via Prolific) to partake in an online survey. We will recruit participants with active X/Twitter usage and ensure a diversity of characteristics, similar to the criteria from Phase 1 recruitment. The exact questions to ask will be written after finalizing our list of hypothesis based on the analysis of Phase 1. However, we can present our preliminary hypothesis with the caveat that it will likely change after Phase 1.

Hypothesis:

- H1: A person’s pre-existing personal opinions, not contextual information around a post, is the biggest predictor regarding their public perception. In other words, someone who agrees with a tweet will often overestimate how many other people agree. Social media context (high engagement metrics) is also a causatory factor in estimates being off, but to a lesser degree.
- H2: A person with stronger opinions is more likely to update beliefs when presented with conflicting information from EchoBreaker. They will either rationalize the gap or outright reject EchoBreaker as unreliable.
- H3: The ability to see EchoBreaker’s synthetic polling data broken down by various demographics helps in building trust. In other words, a person is more likely to believe that EchoBreaker is more accurate with demographic breakdowns. As a result, they are also more likely to update their beliefs.
- H4: People who already have more trust in polling data are more likely to be receptive to data from EchoBreaker.

B Expected Contributions

This research paper will have four main contributions to the researcher community.

Contribution 1: Analysis of what predicts the magnitude in perception gaps at the moment one is browsing a particular social media post.

Contribution 2: Analysis of how people respond when presented with data that conflict with their perception on public opinion. Do they update their beliefs? Conclude inaccuracy of the information or tool presented? Do they rationalize the discrepancies? The answer might be a combination of reasons.

Contribution 3: Design guidelines based on our findings on what makes an effective opinion calibration tool. How should information be presented? Should it be more active or passive? What builds or erodes trust?

Contribution 4: EchoBreaker itself as a research instrument that others can build from. EchoBreaker's repository will be made public.